

Index

1. Overview
 - What is Runebrace
 - Why Runebrace
 - Getting Help
2. Getting Started
 - Downloading and Installing
 - UI Overview
 - Your First Project
 - Understanding File Types
3. Supporting Models
 - Basic workflow
 - Placing supports
 - Features
 -
 - Presets
4. Working with Meshes
 - Repairs
 - Replacing a Mesh/Model
 - Cutting a Mesh
5. Hollowing
6. Printing and Slicing

Glossary

See the Glossary for a list of commonly used terms and definitions.

Overview

Getting Started

Downloading and Installing

OS/Platform Differences

- Keyboard shortcuts

UI Overview

Runebrace's User Interface (UI) is shown below with the major functional areas called out.

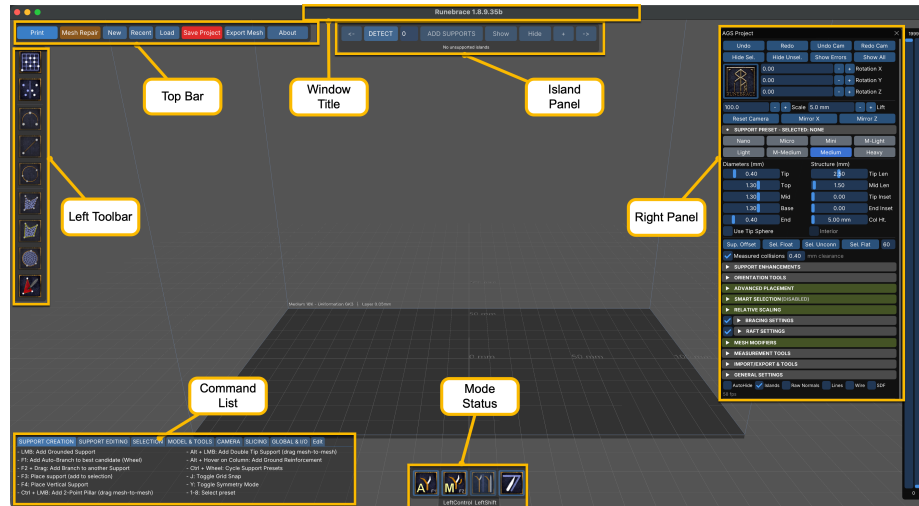


Figure 1: Runebrace's main UI

It's important to get used to the idea that Runebrace has no tooltips anywhere. This is intentional, as tooltips cover up the controls underneath. Controls clearly state their function on the control itself or nearby in the panel.

Functional Areas and Commands

Below you can find short descriptions of each feature.

Top Bar

Print Opens the export window (see Printing and Slicing

Mesh Repair Opens the repair window (see Mesh Repair)

[New] Creates a new empty scene

[Revert] Reloads the file from the last saved version, discarding changes

[Load] Opens a project or a model, with a Recent list

[Save Project] Saves in place; a menu also offers Save Project As

[Export Mesh] Opens the mesh export menu (see Exporting Meshes)

[About] Displays the Runebrace version, the graphics card in use, important links, and a donation button

There are two styles of file bar. Global Settings has a switch, "Use Standard File Buttons", that chooses between them.

Left Toolbar

This toolbar displays icons/buttons for different support tools.



Figure 2: The Left Toolbar

Right Panel

The Right Panel holds features and controls for nearly all of Runebrace's editing functionality.

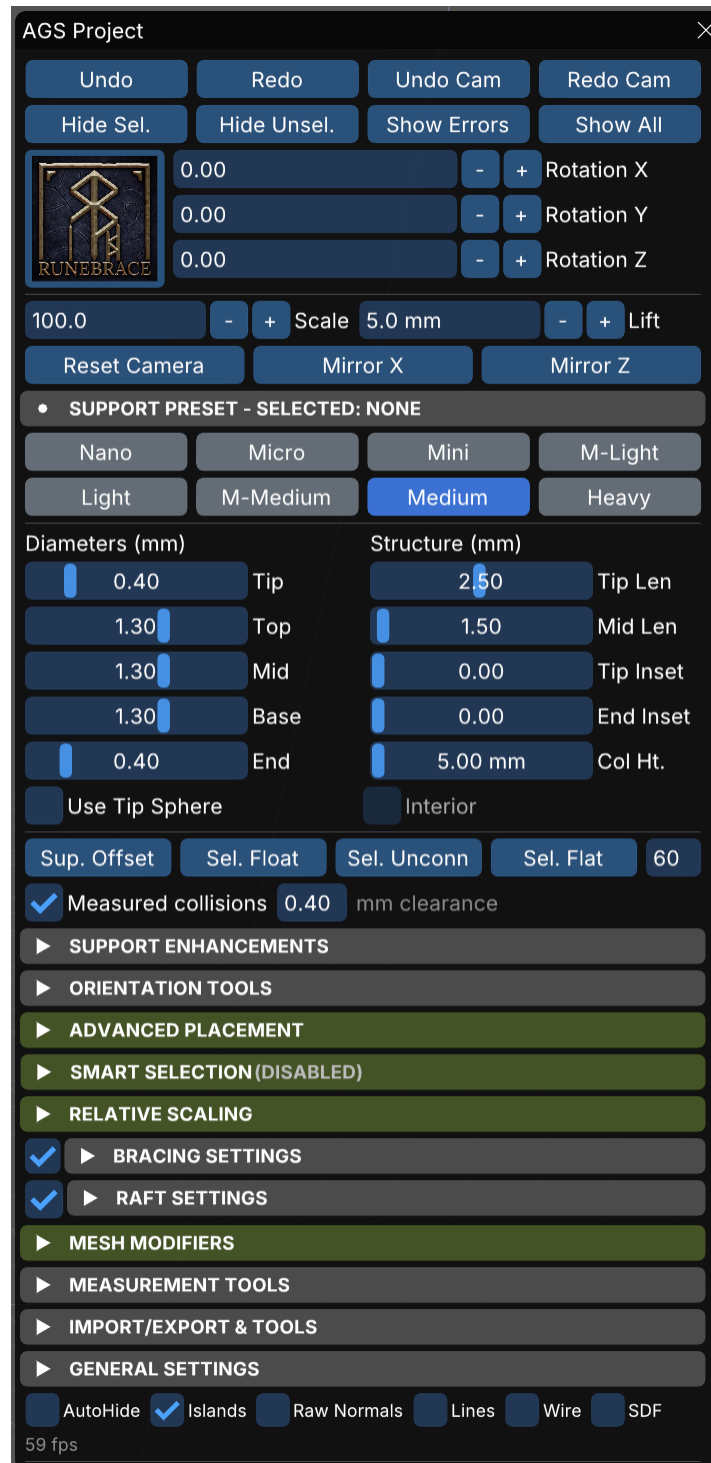


Figure 3: The Right Panel

Most sections in the right panel are collapsible. Each area will be discussed later in this guide.

Command List

The Command List shows what mouse and keyboard actions are currently available. These change with the active tool.

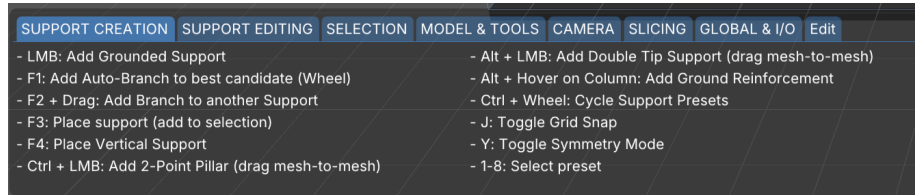


Figure 4: The Command List

The Edit tab allows you to remap some commands to better suit your style and workflow.

You can also find a list of keyboard and mouse shortcuts on the right panel under General Settings.

Mode Status

At the bottom are four icons that display current support modes.



Figure 5: The Mode Status display

- Branch to nearest candidate. Wheel to scroll through candidates.
- Manually add branch via dragging

- NEED CONTENT
- NEED CONTENT

Island Panel

The Island Panel allows you to detect, support, hide, and otherwise interact with islands.

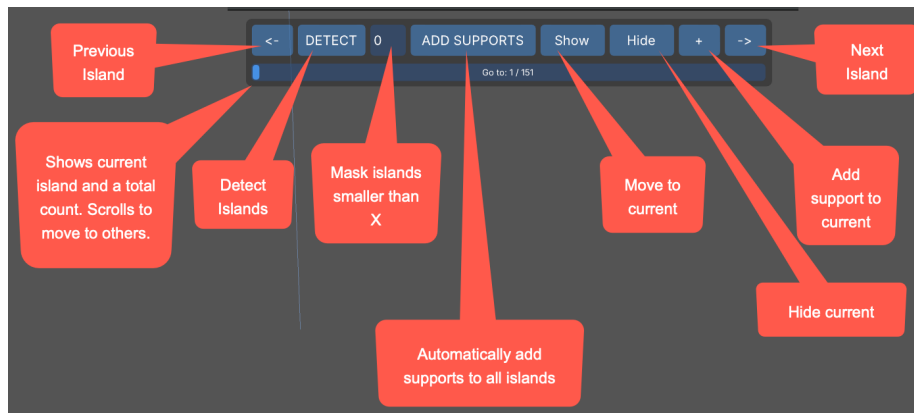


Figure 6: The Island Panel

Controls on this panel are:

- [<-] Move to previous island
- [Detect] Detect islands on the model
- [Island Filter] Filters islands below this size in pixels from being displayed, counted, or resolved by the “Add Supports” command. **Islands are not eliminated, just hidden.**
- [Add Supports] Automatically add supports to islands. Some may be skipped due to mesh complexities.
- [Show] Rotate model to show the current island
- [Hide] Hide the current island from the display and count
- [+] Add a support to the current island
- [->] Move to next island

The scroll area below the buttons displays a current count of the number of islands, as well as an index of which island you’re currently working on. Next and Previous increment/decrement that count as expected. You can also drag the scroll marker to move quickly through supports.

WINDOW TITLE Runebrace reports the outcome of an action in the window title, not in a popup. “Holes applied”, “31 supports shown”, “Model swapped”, errors, and timings all appear there. A reader of the manual should be told to look at it, because several actions give no other feedback.

A Note on Display Scaling

Runebrace's entire UI scales with the window width, starting from a 3840 pixel reference. I.e., a 3840-wide window renders at a scale of 1.0, while a window 1280 wide renders at one third.

Fonts and every aspect of the display scales as well, with nothing being tied to the display DPI.

If you work in a small window, raise the UI Scale and Font Scale settings under Right Panel => General Settings => Global Settings. Both of these multiply on top of the automatic factor.

Your First Project

Understanding File Types

Supporting Models

Supporting Overview

Basic Workflow

CONTENT NEEDED

Features

Support Basics

A support is a chain of control points with a radius at each one: the tip that touches the model, the top of the shaft, the middle, and the base on the plate. Between them are cones. Every diameter is editable.

Support Variants

Grounded The support's tip is on the model, and the base is on the plate

Branching The support joins another support instead of reaching the plate

Column A thick pillar with its own end diameter and height

Double Tip Connects two points of the model, mesh-to-mesh, with no base

Reinforcement A ground reinforcement added to an existing brace

Bracing # : Lateral support between the mid-sections of supports

CONTENT NEEDED

- Horizontal line of layers

Manual Supports

CONTENT NEEDED

Left Toolbar (REWORD)

The left toolbar controls many support options.



Figure 7: Left toolbar icons

Symmetry

CONTENT NEEDED

Path Supports

Path supports are created on a line between two or more supports. A path is drawn from the tip of each support in the order they were selected. (Use F3 + LMB when selecting multiple supports.)

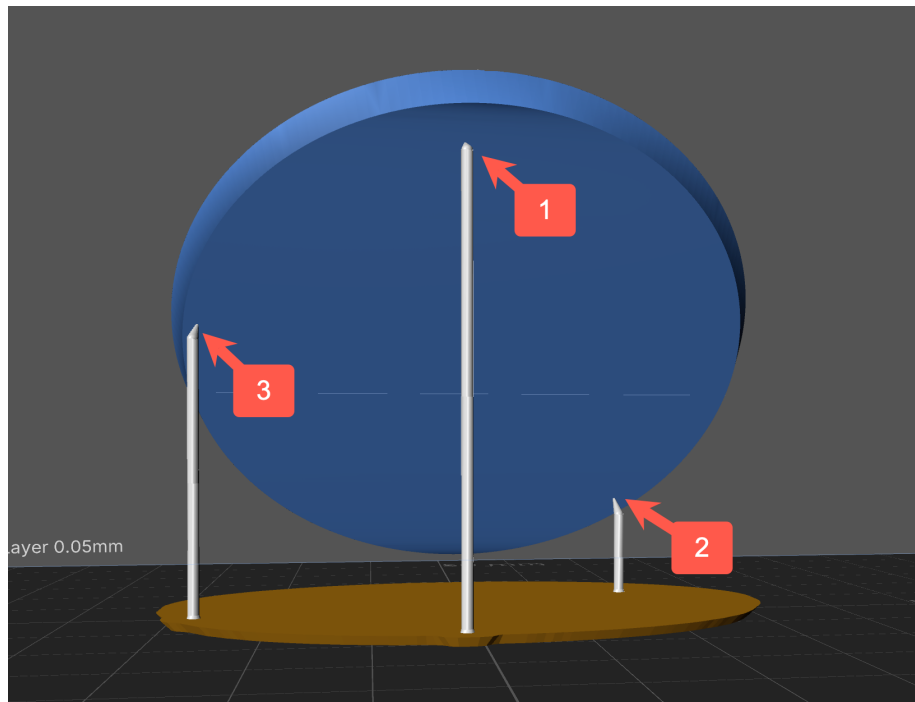


Figure 8: Round Base with Three Supports

Settings controlling spacing of supports for the following features are located in the right panel under Advanced Placement => Line and Fill.

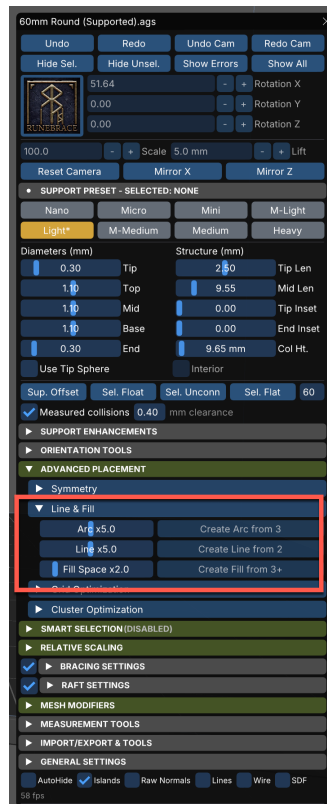


Figure 9: right panel with line and fill highlighted

Important Considerations

Settings in this section are not distances in millimeters; they are multipliers of the support diameter. Runebrace determines how many supports fit in the space between the selected supports, then spreads generated supports evenly along the segment.

Very small values stop having an effect at some point, and are ignored.

A segment too short to fit anything gets no supports.

Line

Adds support along a line between two and only two supports.

Line placement is available through the right panel and the left toolbar (fig. 7).

Spacing for supports along the line is controlled by the right panel under Advanced Placement => Line & Fill => Line.

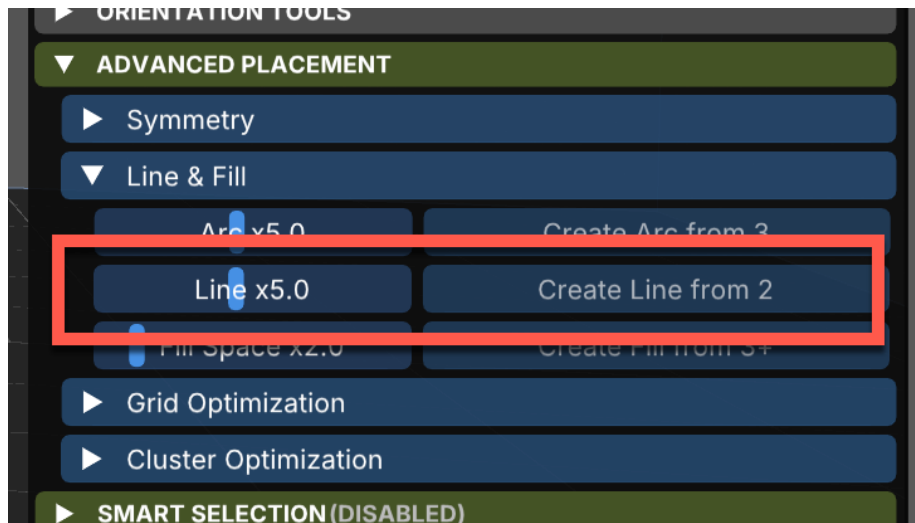


Figure 10: Menu for adjusting Polygon Fill spacing

Selecting support #1, then #2, then creating Line supports results in this:

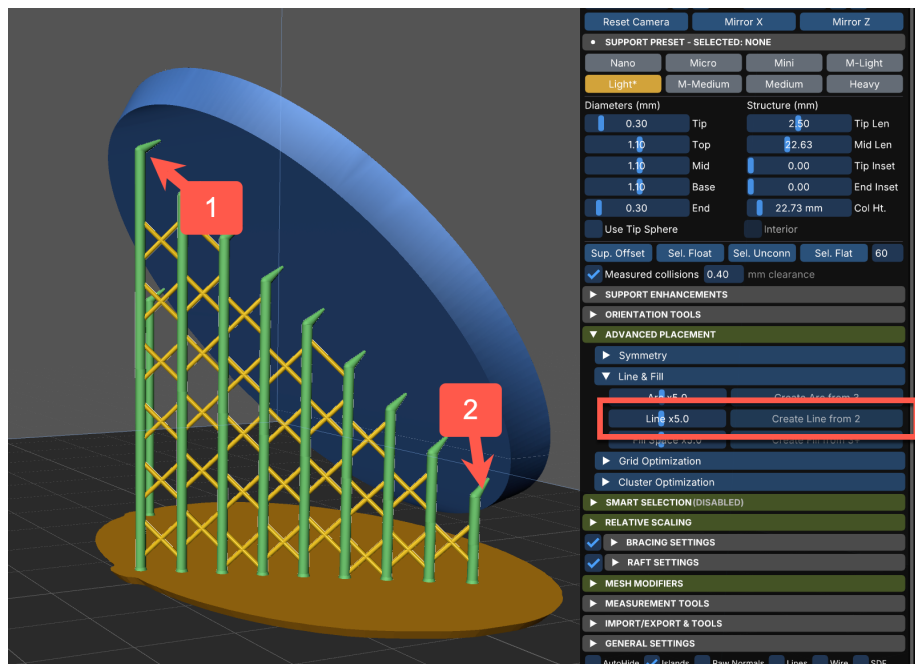


Figure 11: Supports created via line

For more details on how lines work, see the deep dive on lines

Arc

Using Arc adds supports along three selected supports—no more, no fewer.

Arc placement is available through the right panel and the left toolbar (fig. 7).

Configure using the settings in the right panel Advanced Placement => Line & Fill => Arc.

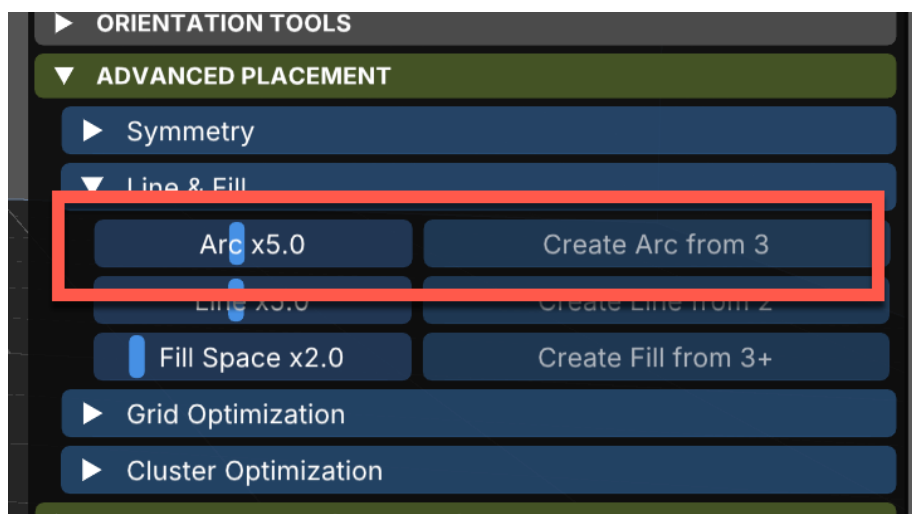


Figure 12: Settings for Arcs

Selecting support #1, then #2, then #3, then creating Arc supports results in this:

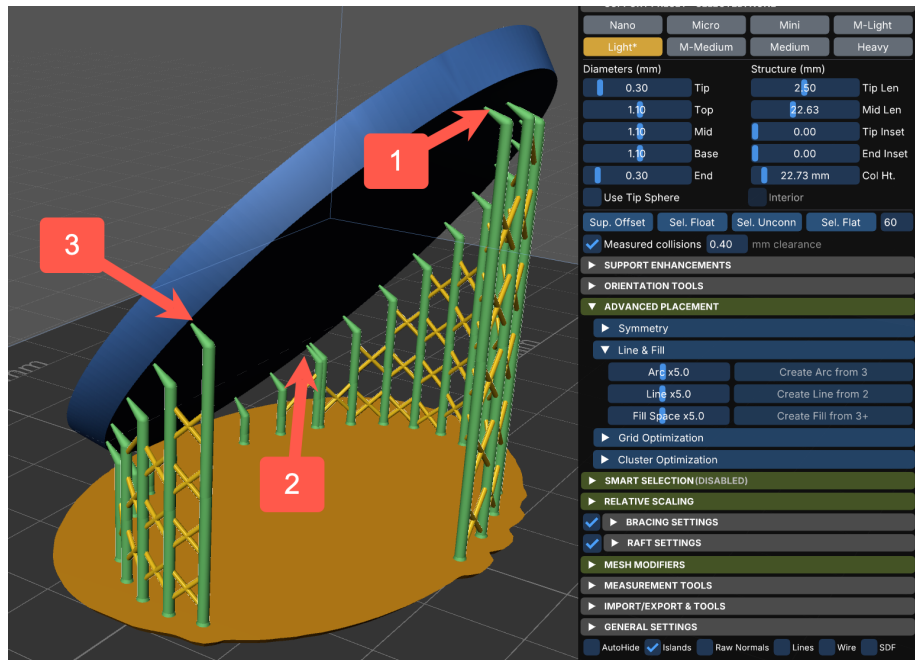


Figure 13: Supports created via Arc

Note the double supports appearing at callout #2. This is due to the spacing along the arc, with one added support being in close proximity to the original #2.

Circle

Using Circle adds supports along a circle path drawn through three supports—no more, no fewer.

Circle placement is available **only** through the left toolbar (fig. 7).

Configure using the settings in the right panel Advanced Placement => Line & Fill => Arc as shown above in fig. 12.

Selecting support #1, then #2, then #3, then creating Circle supports results in this:

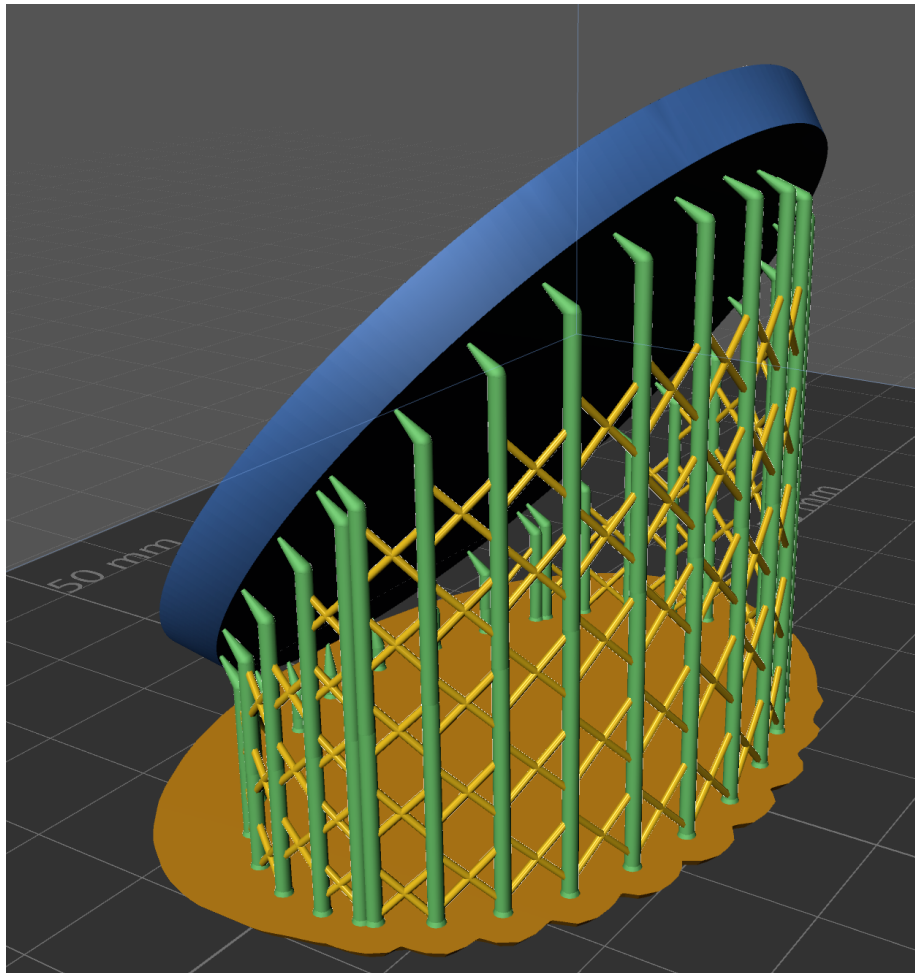


Figure 14: Supports created via Circle

Again, note the doubled supports, this time at #2 and #3.

Fill Basics

Runebrace also provides area fill capabilities.

We'll start with a basic example, then show a more complex, practical example after. Triangle Painting and fills are covered separately below.

Each feature below will show results starting from the above figure fig. 8, a round base with three supports selected in the order they're labeled.

Polygon Fill

Polygon Fill, commonly referred to as Poly Fill, fills an area defined by three or more supports. No rim support is created.

Poly Fill is available through the Fill Space option in the right panel and the corresponding button on the left toolbar (fig. 7).

You can also use the Fill Space button from the right panel (see below).

Using Poly Fill/Fill Space from our starting point results in this:

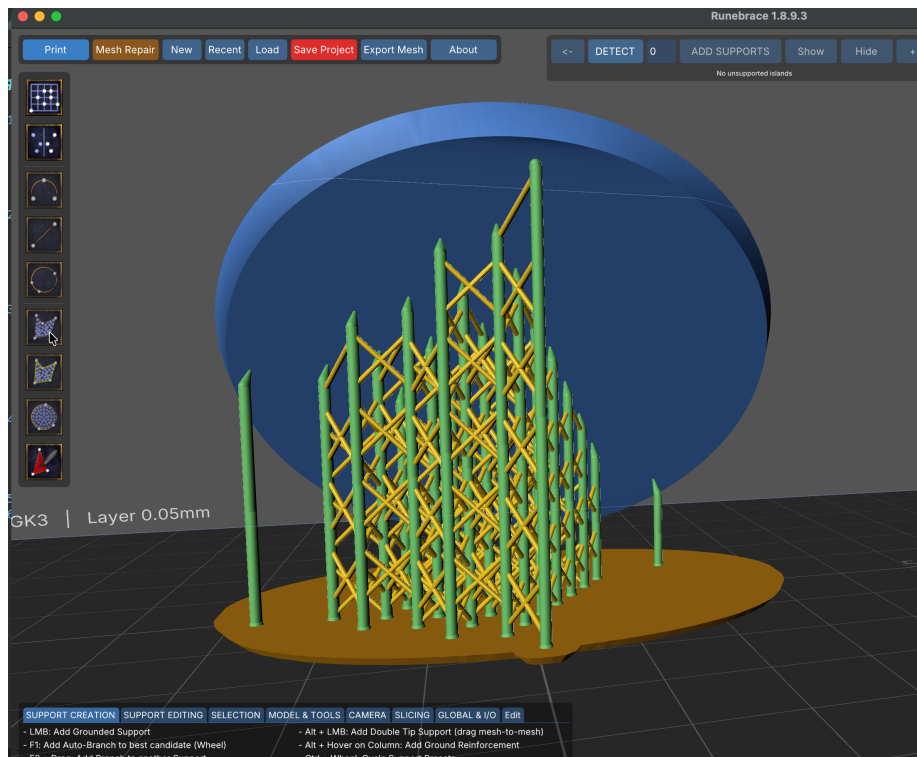


Figure 15: Poly fill

Why: EDIT: ADD USE CASE

Adjust the spacing of the filled supports under right panel => Advanced Placement => Line and Fill => Fill Space



Figure 16: Menu for adjusting Polygon Fill spacing

Perimeter and Fill

Perimeter & Fill does the same as Poly Fill but also lays supports along the area's bordering path itself. Note the menu button's icon shows this via the yellow outline.

Poly Fill is available **only** through the corresponding button on the left toolbar (fig. 7).

Select three or more supports, then press the Perimeter and Fill button (there's no alternative from the right panel). One action creates the perimeter and inside area supports.

Using Perimeter and Fill from our starting point results in this:

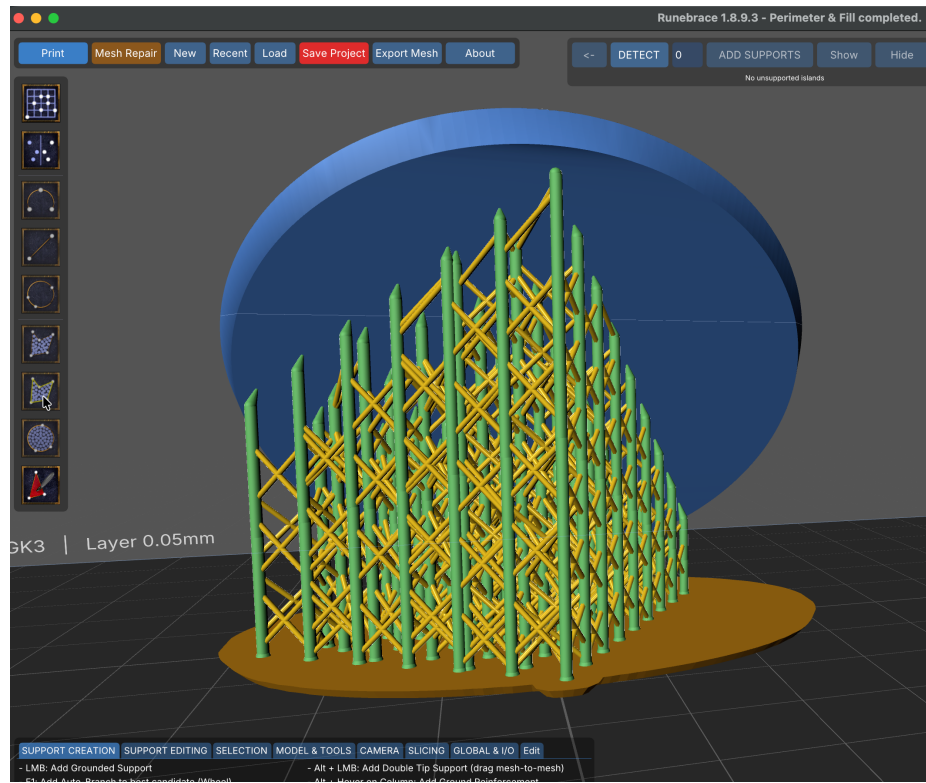


Figure 17: Poly fill with Perimeter

Perimeter and Fill has two separate adjustments:

- **Line:** Spacing between supports on the lines of the outer perimeter
- **Fill Space:** Spacing between supports in the grid that fills the area

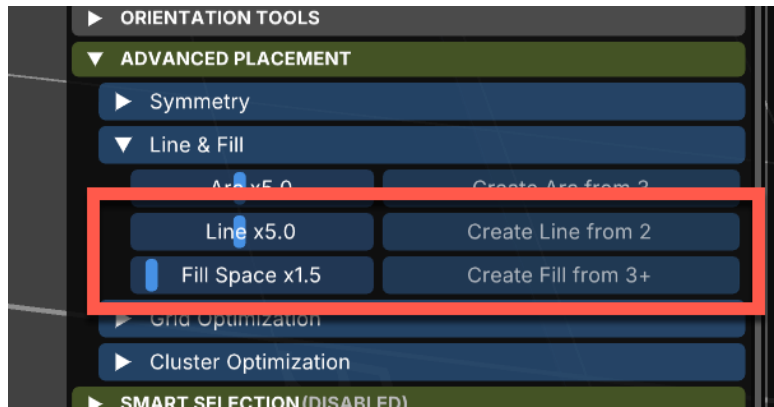


Figure 18: Menu for adjusting both Polygon Fill and Line spacing

These settings are demonstrated in the example below. The perimeter supports are denoted by the red hash lines and were configured with a spacing of 5.0. The inside area was configured with a spacing of 1.5.

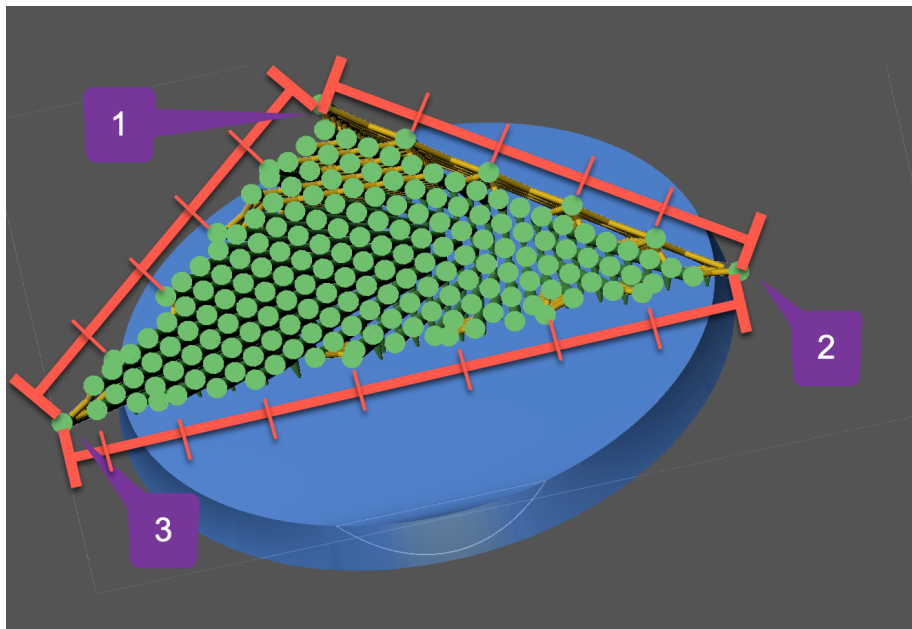


Figure 19: Poly fill with perimeter as seen from underneath

Circle Fill

Circle Fill takes exactly three supports, builds a circular path that passes through their contact points, and fills that. No path supports are created.

Circle Fill is available **only** through the corresponding button on the left toolbar (fig. 7).

Using Circle Fill from our starting point results in this:

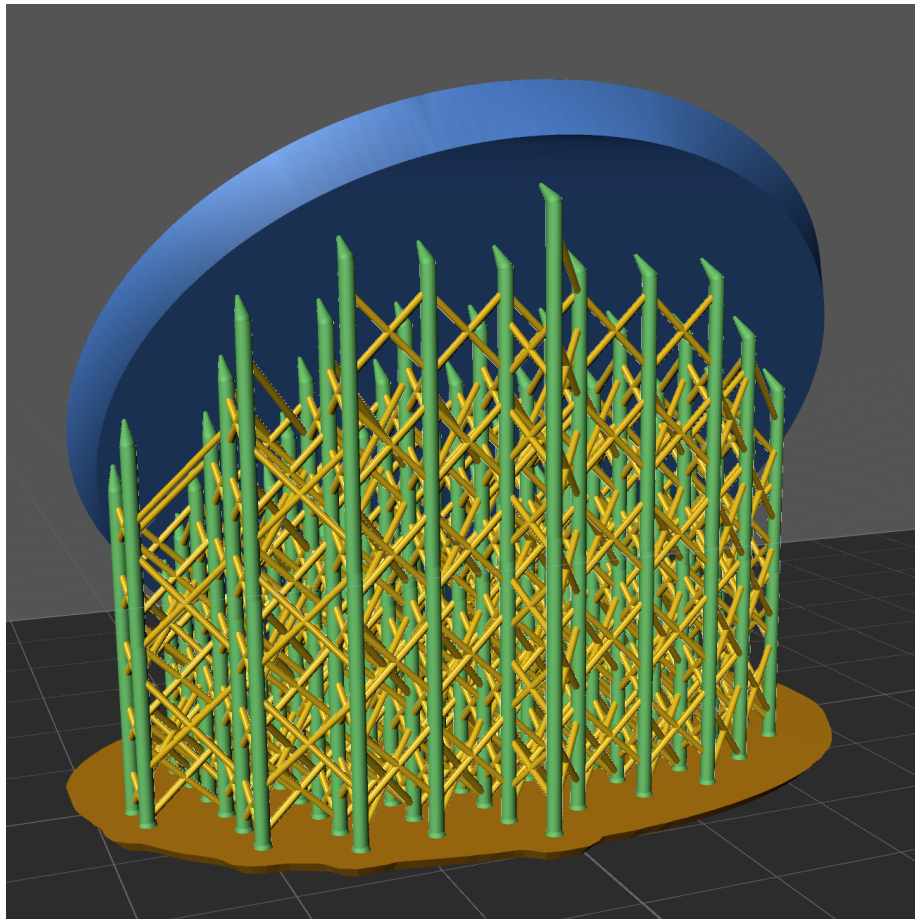


Figure 20: Circle fill

From below, the result looks like:

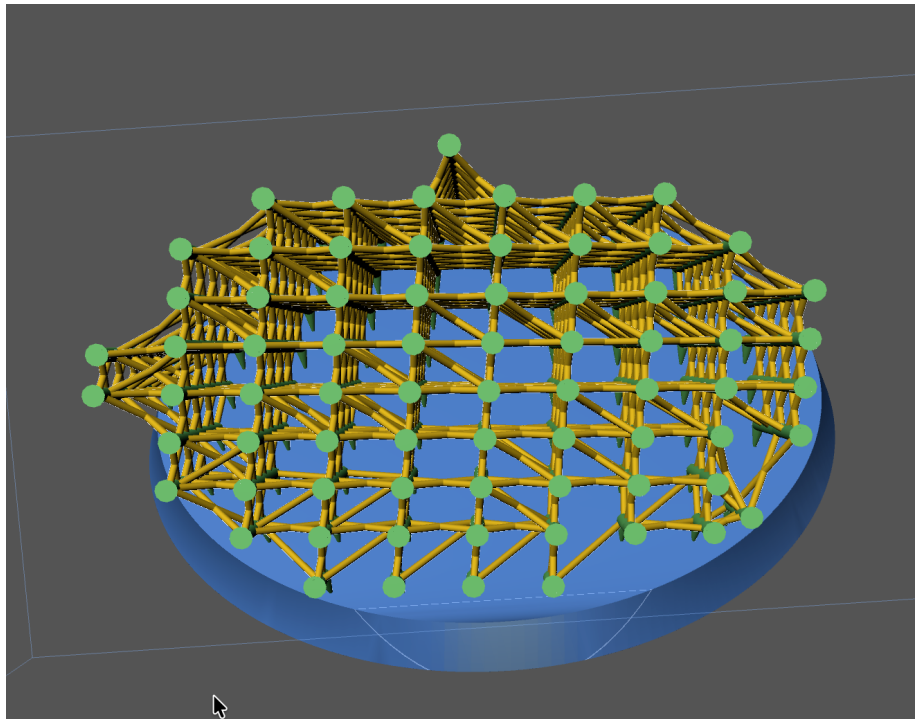


Figure 21: Circle fill viewed from below

Triangle Painting

Rim

Deeper Dive on Placement

The sections below provide a more advanced discussion of how specific placement features work.

More About Lines

Drawing support lines or arcs actually simulates clicking on various points of the line. Those points depend on the distance set in the menu options. Runebrace then performs a raycast on the model.

If the raycast hits the geometry, then a check is made on the impacted face at that point. If the face is not parallel to the build plate, or tilted upward, then a support is placed at that point. **CONTENT NEEDED: why not parallel or upward?**

Areas where the raycast misses the model mean a support can't be placed there; there's no intersecting geometry. Supports are skipped for those areas.

Put differently: It's not a matter of whether the surface is flat. The geometry must be under the ray and must be a valid point for a support.

More About Perimeters

Selection Order Impact on Perimeters and Areas

Perimeters are always created in selection order. The loop is closed from the last support back to the first. This matters greatly as to how supports will be created.

In this first example, supports were selected in order going counter-clockwise.

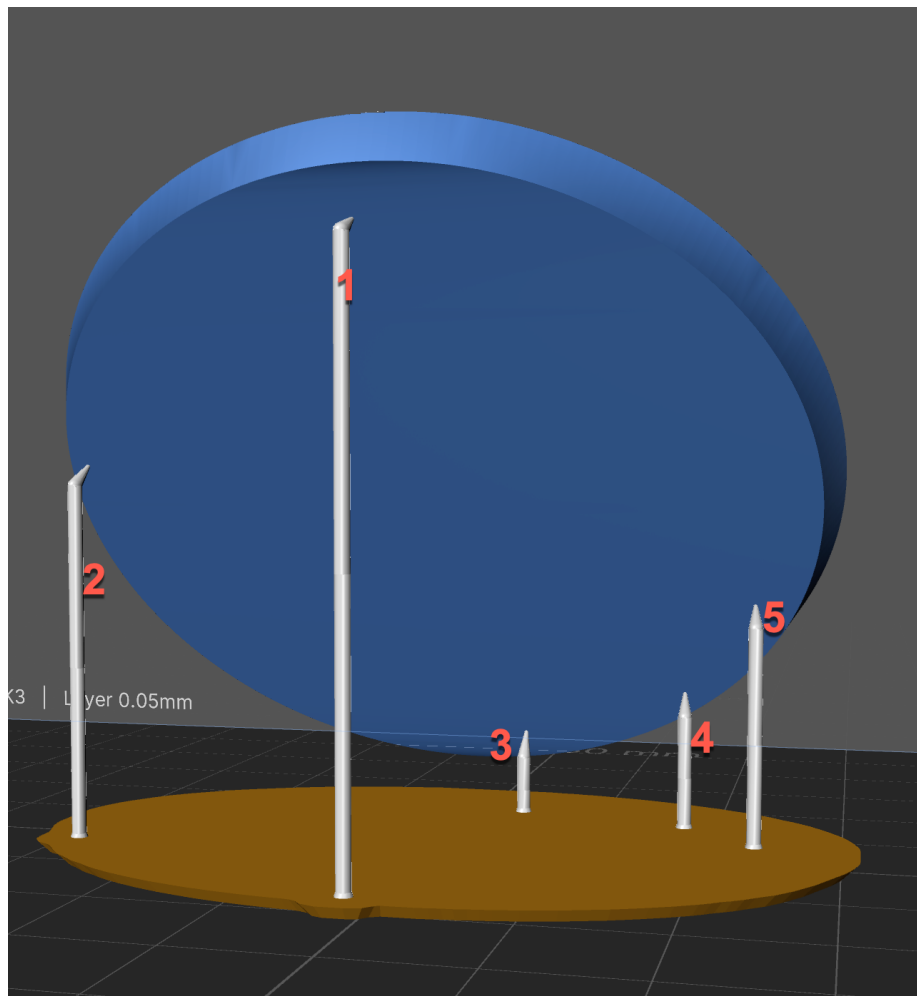


Figure 22: Supports selected in order

Using Perimeter and Fill, the result is

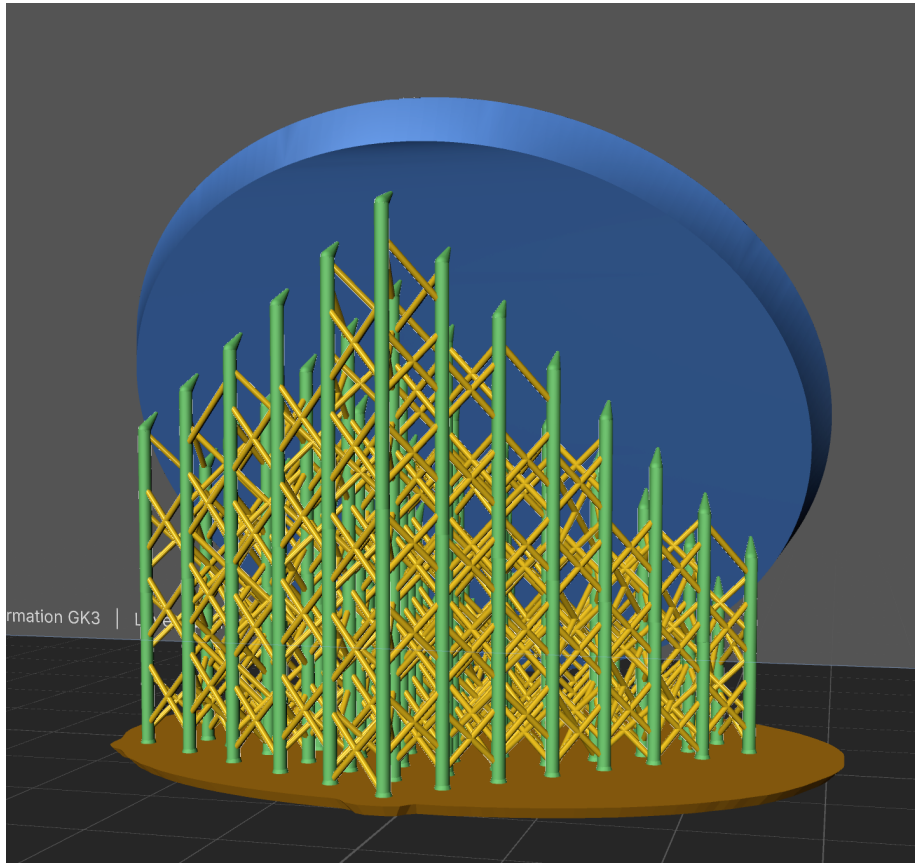


Figure 23: Supports selected in order

Now look at an example with a non-sequential selection:

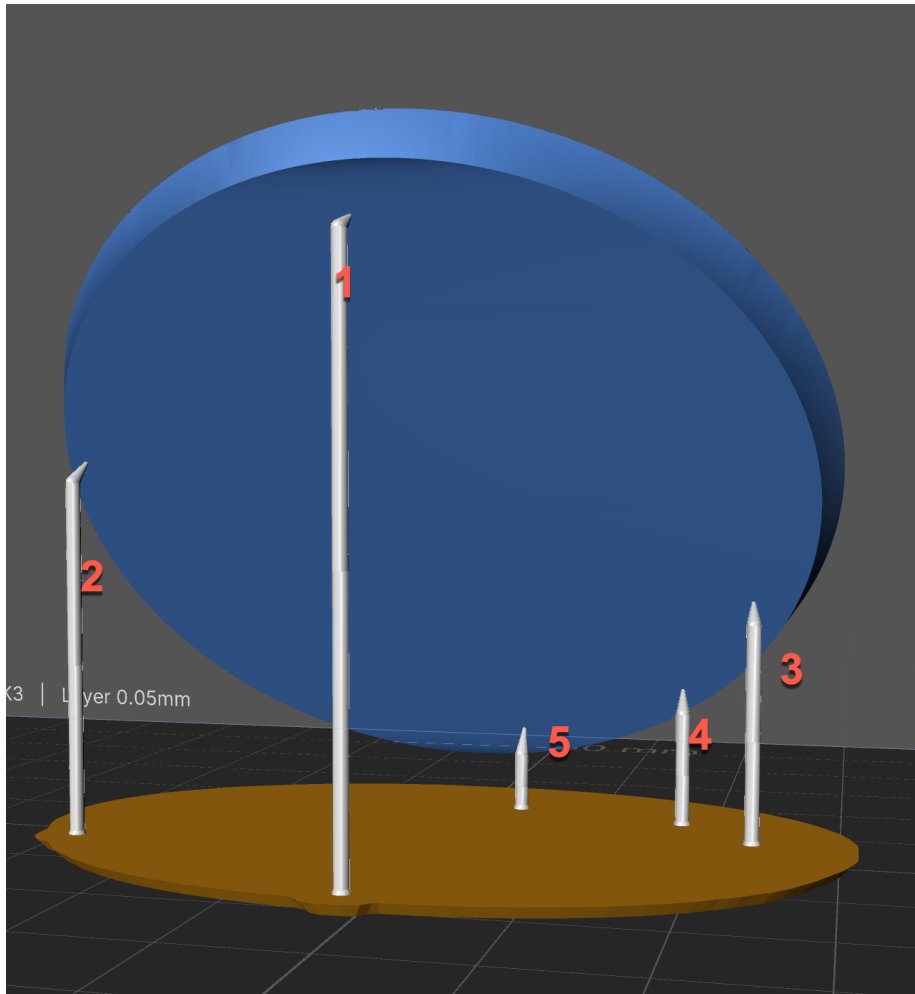


Figure 24: Supports selected in order

Using Perimeter and Fill, the result is

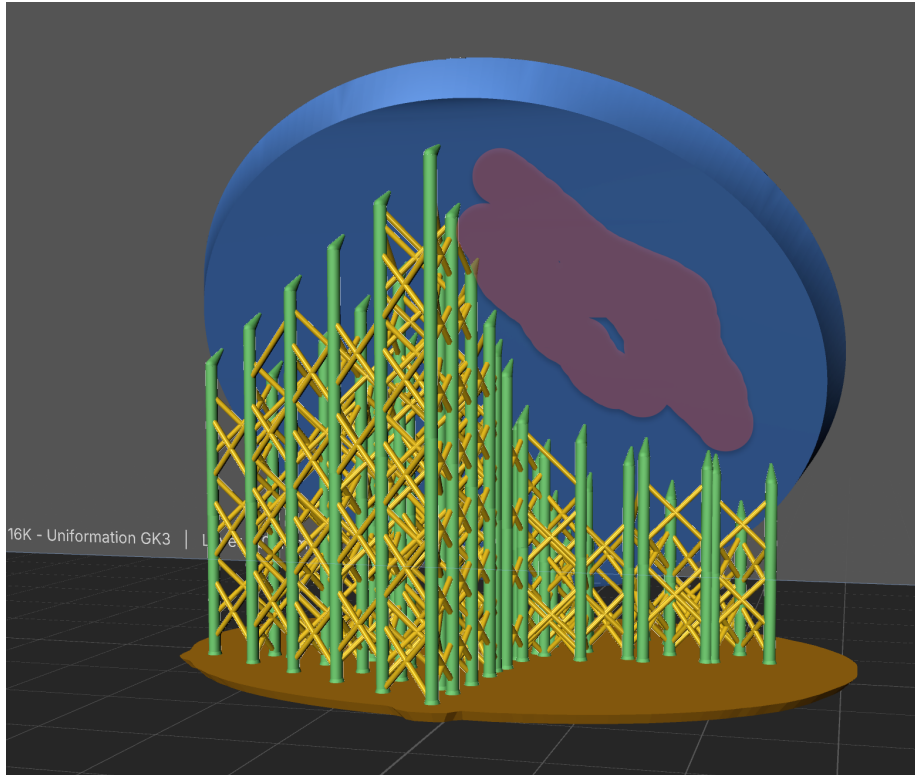


Figure 25: Supports selected in order

It's easy to see the area between supports 1 and 3, marked with the light red scribble, got no supports. This is because the rim path didn't connect those supports, so there wasn't an area to fill.

Understanding Support Impacts on Fill Planes

Area fills are performed on a planar basis. The plane is defined by the first three selected supports, not all selected supports. Additional supports are flattened onto the plane defined by the first.

This results in selected areas not filling the way one might expect.

This example shows a cloak with a surface that significantly changes orientation. Five supports are selected as noted.

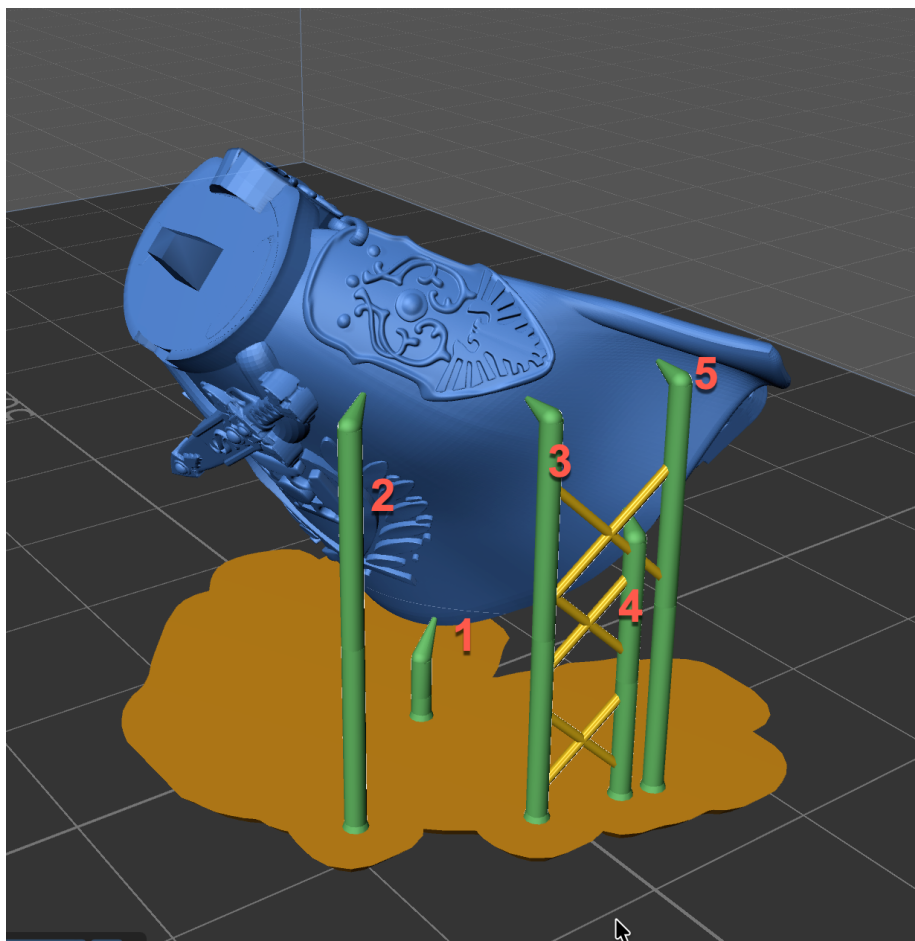


Figure 26: Model with complex surface orientation and five supports

The Poly Fill results in a gap being left, despite that area being valid for additional supports as shown by the ability to place manual supports there—the white transparent support by the mouse cursor.

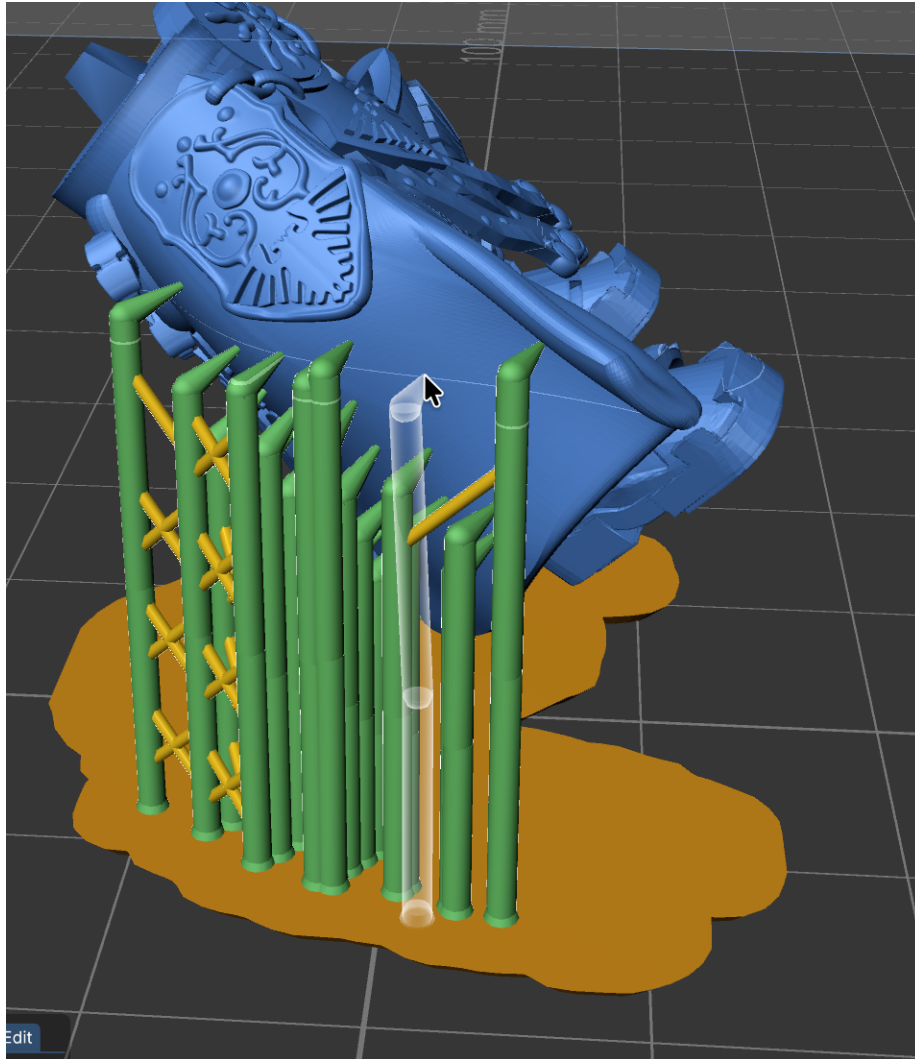


Figure 27: Results with gaps after a fill

Runebrace creates a grid covering the bounding box of the polygon *in the plane defined by the first three supports*. That grid has two axes: the first defined by the line from the first support to the second, and the second axis perpendicular to the first. See the example fig. 19 from earlier for a good visualization of this.

All the selected supports have their tip directions averaged together. Then, each grid point is projected onto the model using that average direction, creating a raycast from each grid point in the average tip direction. This is used to help the fill follow the surface of the defined area.

Each resulting raycast is used to check for intersections with the mesh for appropriate supportable surfaces, and if so, a support is placed. If the ray misses the mesh, then no support is placed, which is why gaps occur in fills.

Understanding Line and Fill Settings

Line and Fill Space settings are multipliers, not distances. Both tie to support diameters for the currently selected support type. Each ties to different parameters: Line ties to support base diameters, while Fill Space ties to a support's Top (not tip!) diameter.

As a result, the same setting on Line and Fill will result in different spacing depending on the targeted support type's configuration.

Undoing Fill Operations

Note that Perimeter and Fill actions are recorded as separate undo operations. This means you'll need to do two separate undo actions to back out that operation.

Math on Determining Placement “step = Base diameter $\times$ Line (or Top diameter $\times$ Fill Space). Gaps = segment length $\div$ step, rounded down. Supports added = gaps $- 1$, spread evenly, so the real spacing is segment length $\div$ gaps. Because the count rounds down, the spacing is always at least the step and can reach almost double it. Yours: $26.73 \text{ mm} \div 4 \text{ gaps} = 6.68 \text{ mm}$, the 6.64 you measured.”

More Complex Examples

You should now understand the basics of creating supports with the various tools. The example above is very simplistic, so let's use more complex

Presets

Working with Meshes

Exporting Meshes

Hollowing

Printing and Slicing

Glossary

Here are common terms and phrases used in Runebrace.

Path Path applies to Line, Arc, and Circle. For two selected supports, Runebrace builds a path using a raycast between two points. For three selected

supports, Runebrace will build a path by creating a curve that passes through all three supports' contact points.

Raycast A virtual ray is cast through the model from a source to a target. This ray can be traced to determine intersection points with the model. For example, when adding line supports, a raycast is made from the tip of the first support to the tip of the second. The ray is examined for where the model is on, above, or below the ray which then determines support placement.

Rim In triangle mode, “Rim” denotes the border of a painted triangle selection. It is determined from the mesh edges, and can have both an outer border *plus* the borders of any holes inside that particular selection. See also: “Outer Rim Only” and “All boundaries” options in the Triangle Paint panel.